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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AIsML)

Literature Survey Report

S.No	Reference no	Year of Publication	Methodology Used	Dataset Used if Any	Performance Metrics	Key Contributions	Limitations
1	[1]	2026	Systematic Review (PRISMA); focused on 6 application categories (ITS, Feedback, Content Gen, etc.).	88 empirical studies filtered from 3,344 records across Scopus, IEEE, and ACM.	76.1% focus on Higher Education; reported academic gains in 45 papers; 54.5% of studies are from Asia.	Synthesized global evidence of LLM impact on academic performance and student engagement.	Lack of longitudinal data; potential positive researcher bias in innovation-focused studies
2	[2]	2026	Design Science Research (DSR); multi-stage heuristic pipeline using semantic clustering and generation.	5 foundational BPM textbooks and Brazilian National Curriculum Standards (BNCC).	Semantic similarity scores: 0.39–0.82; Expert Content Validity Index (I-CVI) > 0.80 for 80% of skills.	Developed a systematic method to derive 35 validated granular skills from high-level professional competencies.	Output quality is strictly bounded by the depth and specificity of the input Content Corpus.
3	[3]	2025	cMALC-D framework; LLM-guided context generation with diversity-based blending (task arithmetic).	3 autonomous traffic signal control environments based on real-world traffic data.	Test rewards: 29.01 (JN 1x3), 172.87 (HZ); consistently outperformed Domain Randomization baselines.	Prevented mode collapse in multi-agent reinforcement learning (MARL) by injecting novel training scenarios.	Current framework not tested on malfunctioning or noisy traffic sensor environments.
4	[4]	2025	EduPlanner; Multiagent system (Evaluator, Optimizer, Analyst) using a novel Skill-Tree student modeling structure.	GSM8K (8.5K problems) and Algebra (222 examples) math datasets.	Overall score of 88 (vs 17 for GPT-3.5 and 49 for zero-shot GPT-4); Depth metric: 88.9.	Introduced the 5-D CIDPP system (Clarity, Integrity, Depth, Practicality, Pertinence) for iterative optimization.	Skill-Tree is subject-specific; multi-agent collaboration results in high LLM resource consumption.
5	[5]	2025	Multi-stage extraction (BERT entity recognition, CRF prediction, and context-	Syllabi from 3 CS courses: Data Structures, OOP, and	Avg F1 entity score: 0.799; Extraction accuracy: 0.824; Avg F1 relation	Automated system prototype for converting unstructured syllabi into visualized	Weak recognition of abstract concepts; difficulty with ambiguous

			aware semantic fusion).	Compiler Principles.	extraction: 0.704.	curriculum knowledge graphs.	cross-chapter semantic relationships.
6	[6]	2025	CurricuLLM; framework using LLMs for high-level subtask planning and reward function code sampling.	Fetch-Slide, Fetch-Push, AntMaze, and Berkeley Humanoid simulation environments.	Humanoid Tracking Error: 0.46 ± 0.38 m/s; Episode Length: 315.44 steps (comparable to expert humans).	Task-level robot curriculum designer; demonstrated successful sim-to-real transfer on physical bipedal hardware.	Success rate is highly dependent on curriculum quality, leading to high variance in experimental results.
7	[7]	2024	IAM-CRBA; iterative architecture using Customized Role-Based Agents (Teacher, Programmer, Sales).	Real course test involving 60 junior undergraduate students from BIT.	Avg quality score: 8.6/10; Code correctness: 7.3/10; Code conciseness: 8.6/10.	Automated generation of educational programming projects without requiring human-in-the-loop intervention.	Small convenience sample size; code reliability decreases as project complexity and learning steps increase.
8	[8]	2024	G-AI instructional design framework grounded in PCK theory and mathematical problem chains.	High school mathematics dataset covering 183 lessons across 21 chapters.	Practical value (V1): 6.085/8; Subject content (C1): 6.680/8; Subject literacy (T2): 6.049/8.	Created a comprehensive tool for assessing LLM pedagogical capability; identified Statistics as a strong module.	Poor performance in mathematical culture and interdisciplinary design (< 4.0/8); PCK-outcome link unproven.
9	[9]	2024	ASSISTRAG; utilizes Curriculum Assistant Learning and Reinforced Preference Optimization (DPO).	HotpotQA, 2WikiMultiHop QA, and Bamboogle complex reasoning datasets.	F1 accuracy: 45.6; Inference time: 5.73s; Operating cost: \$0.009/question.	Integrated an intelligent plugin for frozen LLMs to manage internal memory and external knowledge retrieval.	Added computational complexity and system latency; performance depends on external knowledge quality.
10	[10]	2024	Human-Centered Design (HCD); iterative 18-month cycle of prototyping, workshops, and testing.	Curriculum coordinators across 5 faculties (Healthcare, Engineering, Architecture, etc.).	Finding info: M=4.2/5; Content review: M=4.1/5; Transferability of results: M=4.3/5.	interactive LLM collaborator mapping course descriptions to UN Sustainable Development Goals (SDGs).	Small survey sample (n=4); pending full integration with student information systems like Peppi.

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Signature of the Team Members

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